

A Spindle-Exact on a gene set S

Query: 12 genes (a contiguous set) 

niche j: S meets 1 of 16 blocks



block 14: 12 of its 16 genes
are in S (filled: $S \times S$)

niche j': S meets 1 of 15 blocks



block 8: 12 of its 24 genes
are in S (filled: $S \times S$)

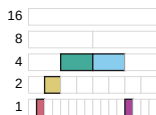
Stored per tile: block logs + block covariances.
Per block, the $S \times S$ part of the stored covariance
is eigendecomposed (stored log if S covers it).

$$d_B^S(q, t) = \sqrt{\frac{\sum_b \log Q_{S \cap b} - \log T_{S \cap b} \|F\|_F^2}{\sqrt{|S|}}}$$

every tile of every niche: exact ranking

2 Interval index (approximate)

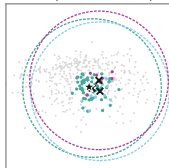
(i) index: every block (here niche j, block 14,
16 genes) $\rightarrow$ dyadic intervals of ≤ 16 genes



genes 1–16 of the block; filled: the query's pieces

per interval of s genes: $\sim 32 \cdot \sqrt{s}$ node means;
every tile within ϵ_s of its node mean
(stored: node means + one code per tile)

tiles on one 4-gene interval (PCA):
× node mean, dashed ϵ ball, ★ query



(ii) query $\rightarrow$ segments (runs in a block)
 $\rightarrow$ fewest dyadic pieces, per niche

niche j: 1 segment, 5 pieces



niche j': 3 segments, 7 pieces



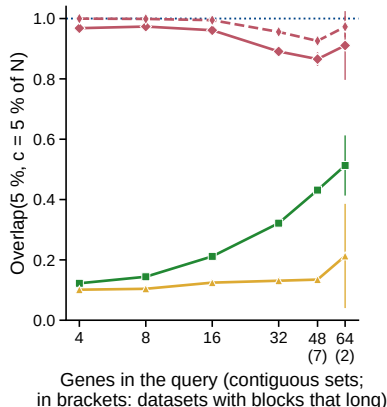
(iii) score every tile t:

$$\hat{d}(t) = \sqrt{\frac{\sum_l \log Q_l - \mu_l(t) \|F\|_F^2}{|S|}}$$

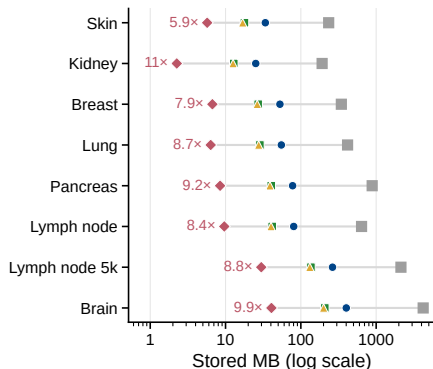
$$|\hat{d}(t) - d_{\text{own}}(t)| \leq \sqrt{\frac{\sum_l \epsilon_l^2}{|S|}} \quad (\text{here: niche j } 2.87, j' 1.96)$$

$\mu_l(t)$: t's node mean on piece l; d_{own} : same sum with t's own logs

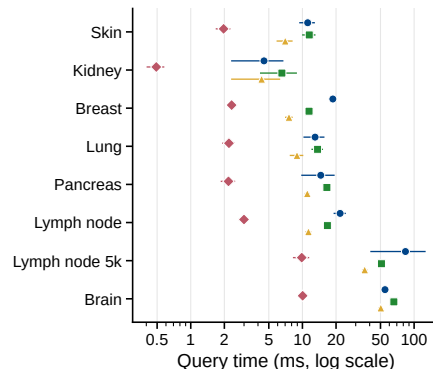
B



C



D



● Spindle-Exact (ground truth) ◆ Interval decomposition, own logs (no index) ▲ Padding + Spindle-Exact
■ Imputation + Spindle-Exact ■ Whole covariances